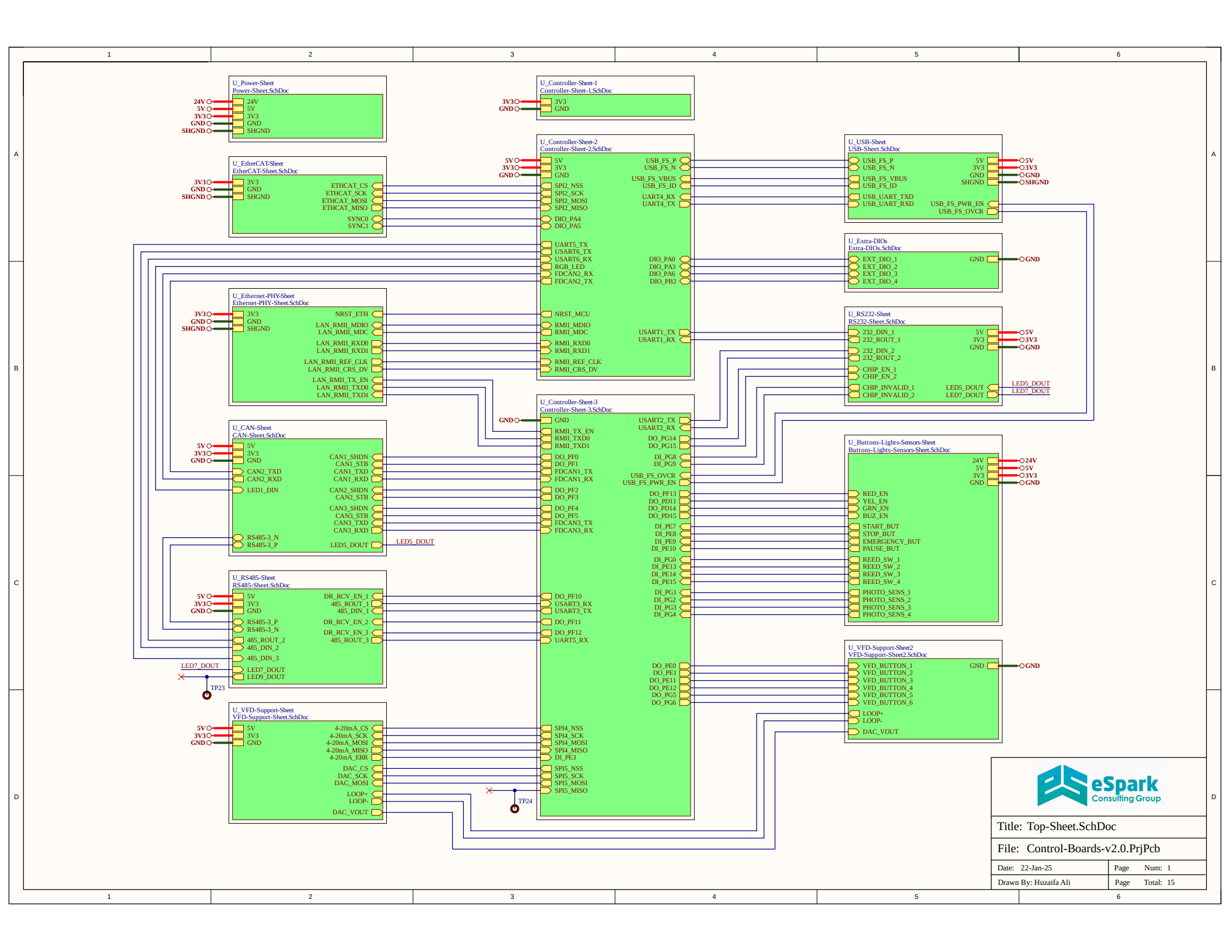
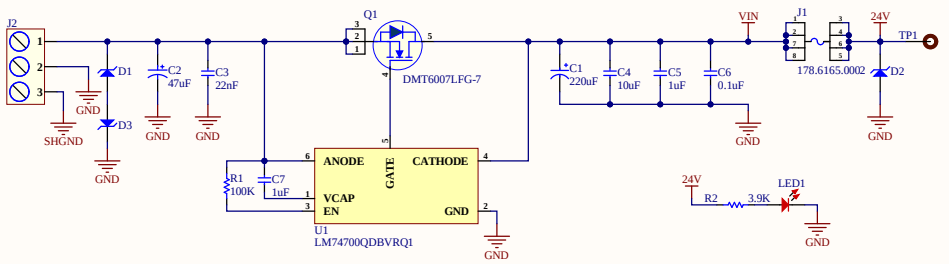
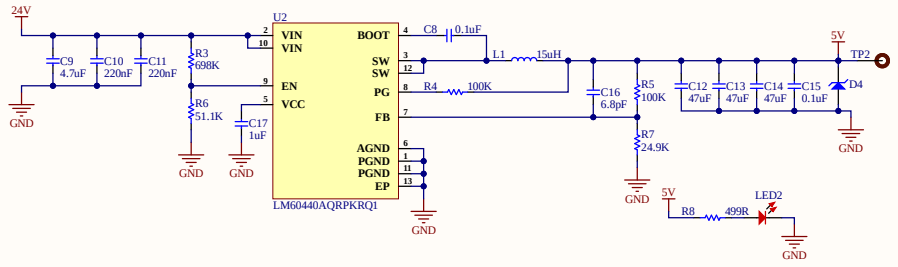

U_Power-Sheet
Power-Sheet.SchDoc
24V
5V
3V3
GND
SHGND
U_Controller-Sheet-1
Controller-Sheet-1.SchDoc
3V3
GND
U_EtherCAT-Sheet
EtherCAT-Sheet.SchDoc
ETHCAT_CS
ETHCAT_SCK
ETHCAT_MOSI
ETHCAT_MISO
SYNC0
SYNC1
U_Controller-Sheet-2
Controller-Sheet-2.SchDoc
SPI2_NSS
SPI2_SCK
SPI2_MOSI
SPI2_MISO
DIO_PA4
DIO_PA5
UART5_TX
USART6_TX
USART6_RX
RGB_LED
FDCAN2_RX
FDCAN2_TX
USB_FS_P
USB_FS_N
USB_FS_VBUS
USB_FS_ID
UART4_RX
UART4_TX
DIO_PA0
DIO_PA3
DIO_PA6
DIO_PB2
U_USB-Sheet
USB-Sheet.SchDoc
USB_FS_P
USB_FS_N
USB_FS_VBUS
USB_FS_ID
USB_UART_TXD
USB_UART_RXD
USB_FS_PWR_EN
USB_FS_OVCR
U_Extra-DIOs
Extra-DIOs.SchDoc
EXT_DIO_1
EXT_DIO_2
EXT_DIO_3
EXT_DIO_4
U_Ethernet-PHY-Sheet
Ethernet-PHY-Sheet.SchDoc
NRST_ETH
LAN_RMII_MDIO
LAN_RMII_MDC
LAN_RMII_RXD0
LAN_RMII_RXD1
LAN_RMII_REF_CLK
LAN_RMII_CRS_DV
LAN_RMII_TX_EN
LAN_RMII_TXD0
LAN_RMII_TXD1
NRST_MCU
RMII_MDIO
RMII_MDC
RMII_RXD0
RMII_RXD1
RMII_REF_CLK
RMII_CRS_DV
U_RS232-Sheet
RS232-Sheet.SchDoc
232_DIN_1
232_ROUT_1
232_DIN_2
232_ROUT_2
CHIP_EN_1
CHIP_EN_2
CHIP_INVALID_1
CHIP_INVALID_2
LED5_DOUT
LED7_DOUT
U_CAN-Sheet
CAN-Sheet.SchDoc
CAN1_SHDN
CAN1_STB
CAN1_TXD
CAN1_RXD
CAN2_TXD
CAN2_RXD
LED1_DIN
CAN2_SHDN
CAN2_STB
CAN3_SHDN
CAN3_STB
CAN3_TXD
CAN3_RXD
RS485-3_N
RS485-3_P
LED5_DOUT
U_Controller-Sheet-3
Controller-Sheet-3.SchDoc
RMII_TX_EN
RMII_TXD0
RMII_TXD1
DO_PF0
DO_PF1
FDCAN1_TX
FDCAN1_RX
DO_PF2
DO_PF3
DO_PF4
DO_PF5
FDCAN3_TX
FDCAN3_RX
USART2_TX
USART2_RX
DO_PG14
DO_PG15
USB_FS_OVCR
USB_FS_PWR_EN
DI_PG8
DI_PG9
DO_PF13
DO_PD11
DO_PD14
DO_PD15
DI_PE7
DI_PE8
DI_PE9
DI_PE10
DI_PG0
DI_PE13
DI_PE14
DI_PE15
DI_PG1
DI_PG2
DI_PG3
DI_PG4
DO_PF10
USART3_RX
USART3_TX
DO_PF11
DO_PF12
UART5_RX
DO_PE0
DO_PE1
DO_PE11
DO_PE12
DO_PG5
DO_PG6
SPI4_NSS
SPI4_SCK
SPI4_MOSI
SPI4_MISO
DI_PE3
SPI5_NSS
SPI5_SCK
SPI5_MOSI
SPI5_MISO
U_Buttons-Lights-Sensors-Sheet
Buttons-Lights-Sensors-Sheet.SchDoc
RED_EN
YEL_EN
GRN_EN
BUZ_EN
START_BUT
STOP_BUT
EMERGENCY_BUT
PAUSE_BUT
REED_SW_1
REED_SW_2
REED_SW_3
REED_SW_4
PHOTO_SENS_1
PHOTO_SENS_2
PHOTO_SENS_3
PHOTO_SENS_4
U_RS485-Sheet
RS485-Sheet.SchDoc
DR_RCV_EN_1
485_ROUT_1
485_DIN_1
DR_RCV_EN_2
DR_RCV_EN_3
485_ROUT_3
485_ROUT_2
485_DIN_2
485_DIN_3
LED7_DOUT
LED9_DOUT
TP23
U_VFD-Support-Sheet2
VFD-Support-Sheet2.SchDoc
VFD_BUTTON_1
VFD_BUTTON_2
VFD_BUTTON_3
VFD_BUTTON_4
VFD_BUTTON_5
VFD_BUTTON_6
LOOP+
LOOP-
DAC_VOUT
U_VFD-Support-Sheet
VFD-Support-Sheet.SchDoc
4-20mA_CS
4-20mA_SCK
4-20mA_MOSI
4-20mA_MISO
4-20mA_ERR
DAC_CS
DAC_SCK
DAC_MOSI
LOOP+
LOOP-
DAC_VOUT
TP24
eSpark
Consulting Group
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File: Control-Boards-v2.0.PrjPcb
Date: 22-Jan-25
Page Num: 1
Drawn By: Huzaifa Ali
Page Total: 15

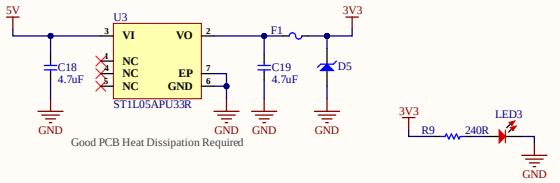
24V INPUT

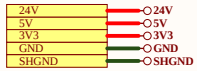


5V BUCK CONVERTER



3.3V LINEAR REGULATOR

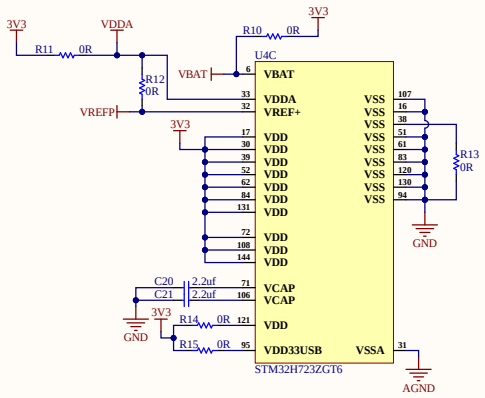


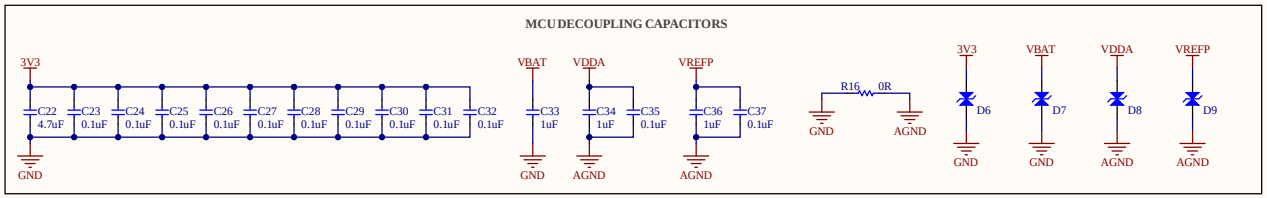


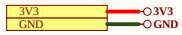
eSpark Consulting Group

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File: Control-Boards-v2.0.PrjPcb	
Date: 22-Jan-25	Page Num: 2
Drawn By: Huzaifa Ali	Page Total: 15

CONTROLLER POWER SHEET

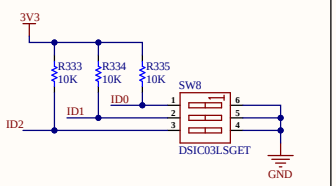


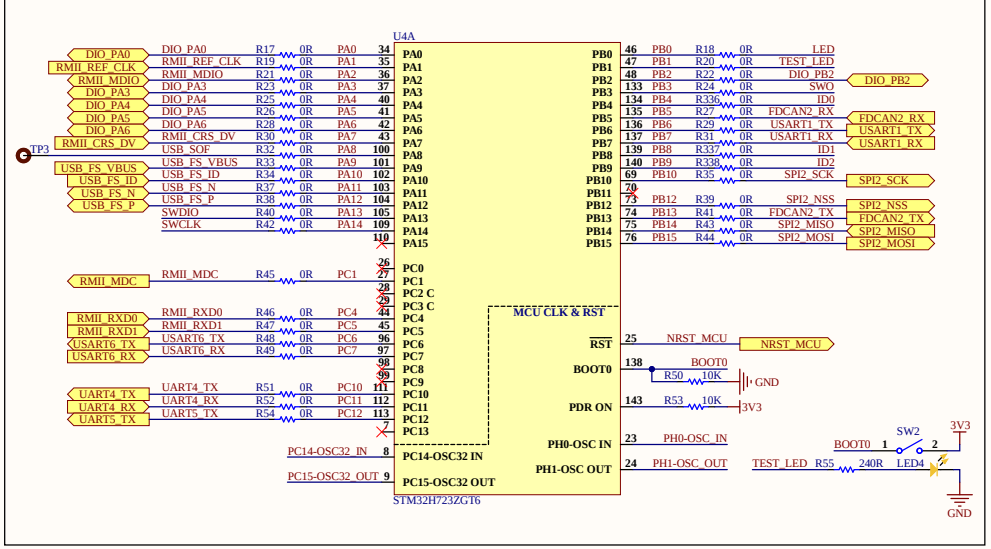


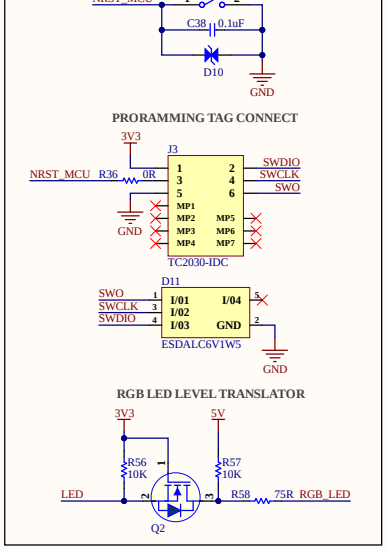


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Date: 22-Jan-25	Page Num: 3
Drawn By: Huzaifa Ali	Page Total: 15







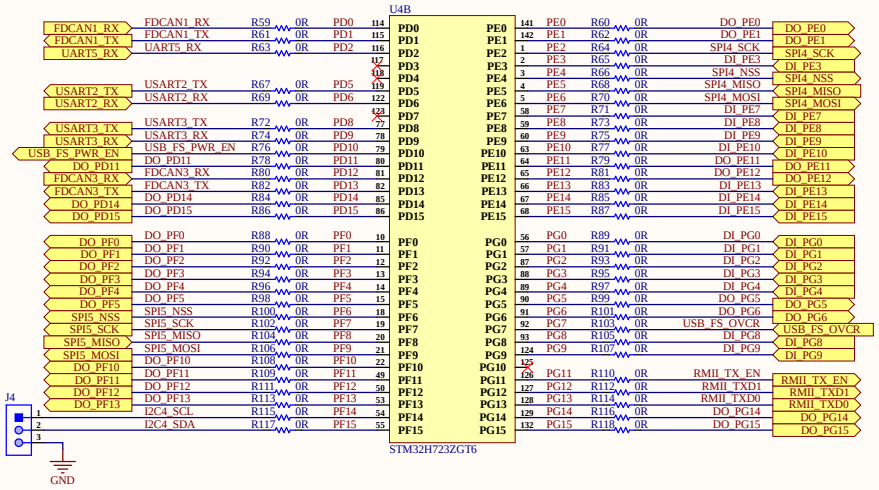
Top-Sheet[3B] RGB_LED RGB_LED

5V 5V
3V3 3V3
GND GND

Title:	Controller-Sheet-2.SchDoc
File:	Control-Boards-v2.0.PrjPcb
Date: 22-Jan-25	Page Num: 4
Drawn By: Huzaifa Ali	Page Total: 15

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CONTROLLER IO SHEET 2

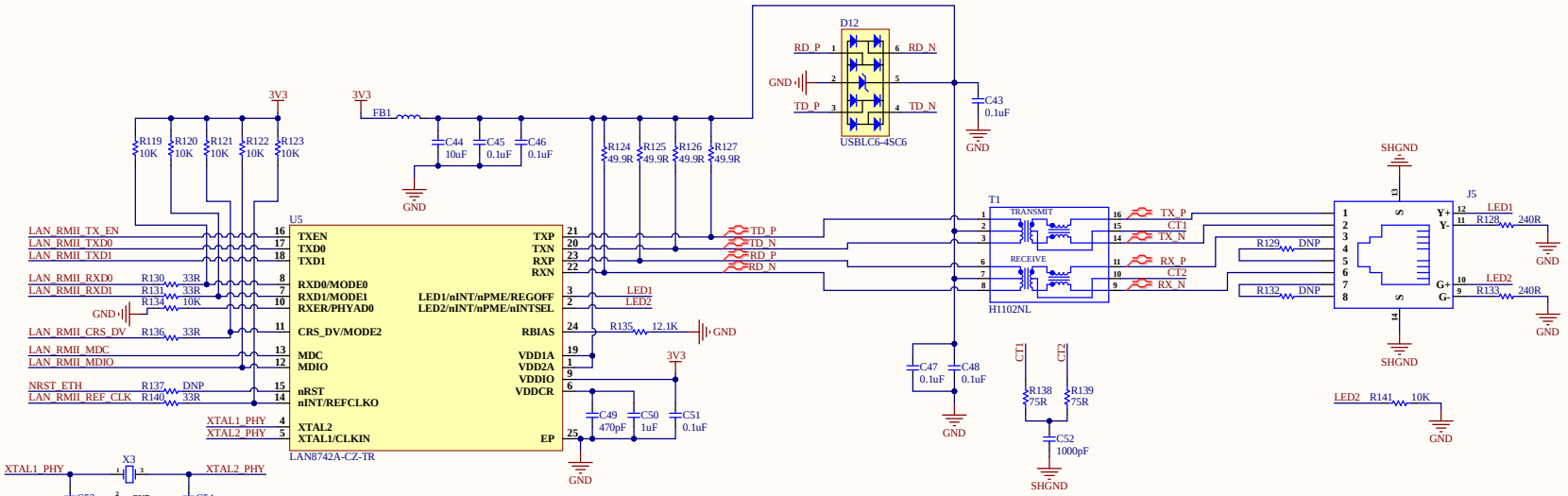


GND ○ GND

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Title:	Controller-Sheet-3.SchDoc
File:	Control-Boards-v2.0.PrjPcb
Date: 22-Jan-25	Page Num: 5
Drawn By: Huzaifa Ali	Page Total: 15

ETHERNET PHY SHEET



NRST_ETH
LAN_RMII_TX_EN
LAN_RMII_TXD0
LAN_RMII_TXD1
LAN_RMII_MDC

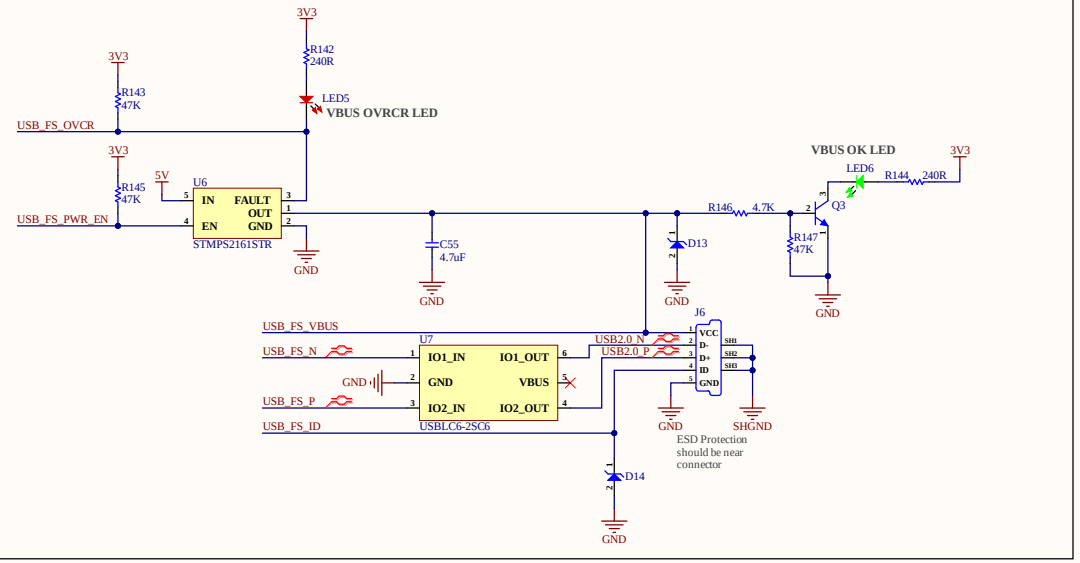
LAN_RMII_RXD0
LAN_RMII_RXD1
LAN_RMII_CRS_DV
LAN_RMII_MDIO
LAN_RMII_REF_CLK

3V3
GND
SHGND

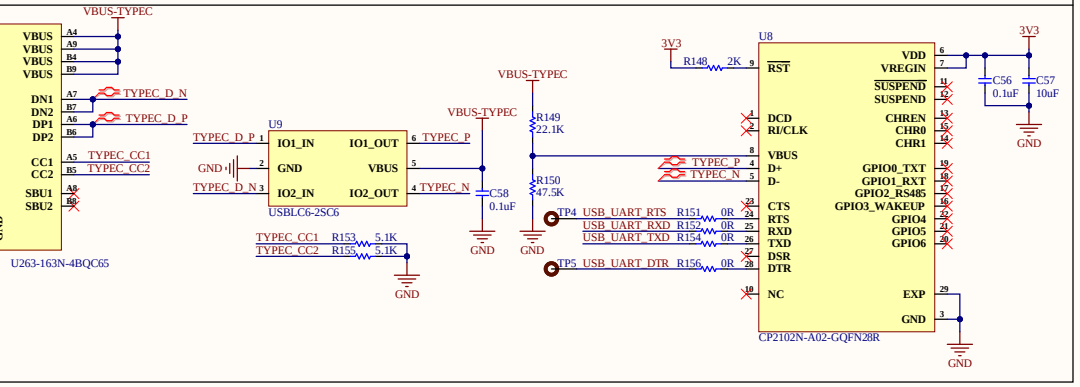
eSpark Consulting Group

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File:	Control-Boards-v2.0.PrjPcb
Date: 22-Jan-25	Page Num: 6
Drawn By: Huzaifa Ali	Page Total: 15

MICRO USB



DEBUG TYPE-C



USB_FS_PWR_EN
USB_FS_OVCR
USB_FS_VBUS
USB_FS_ID
USB_FS_N
USB_FS_P

USB_UART_RXD
USB_UART_TXD

5V
3V3
GND
SHGND

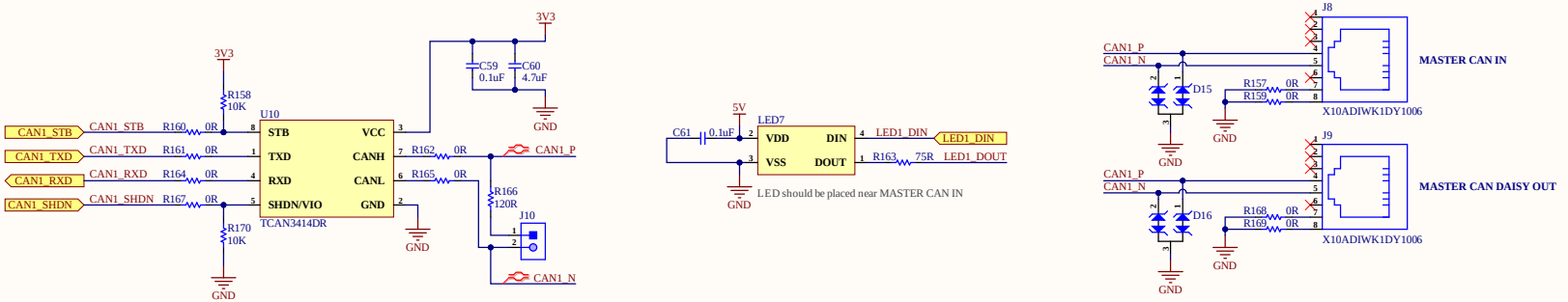
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Title: USB-Sheet.SchDoc

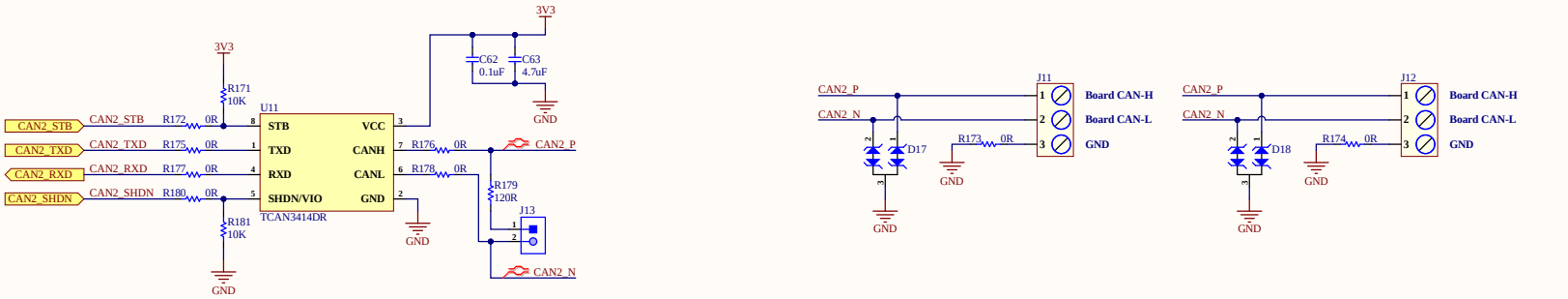
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Date: 29-Jan-25	Page Num: 7
Drawn By: Huzaifa Ali	Page Total: 15

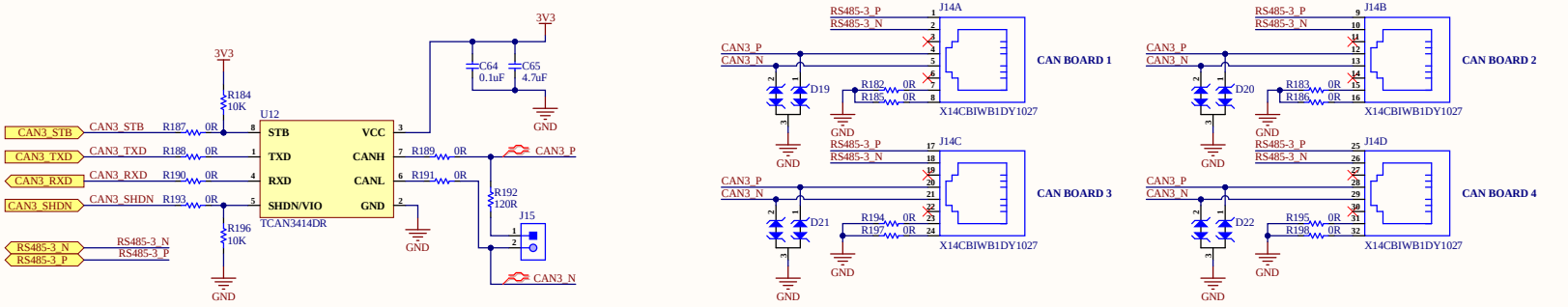
CAN 1



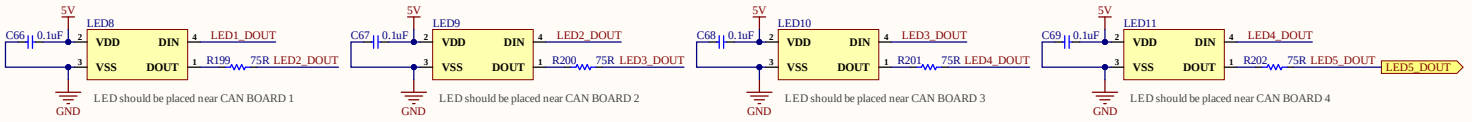
CAN 2



CAN 3



LEDs

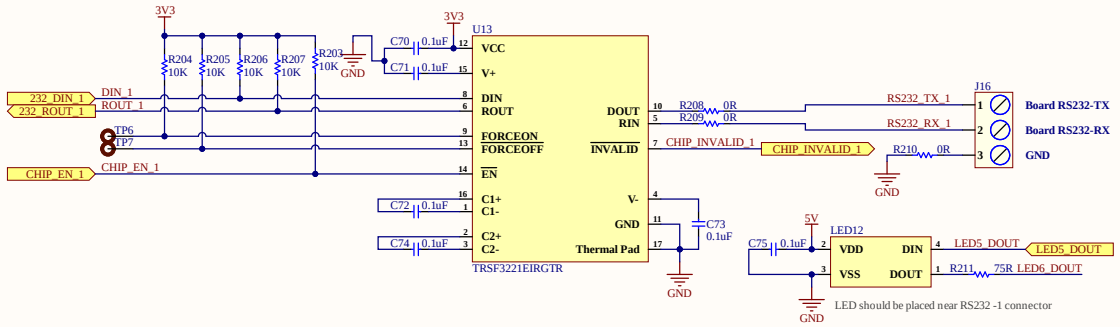


5V
3V3
5V
3V3
GND
GND

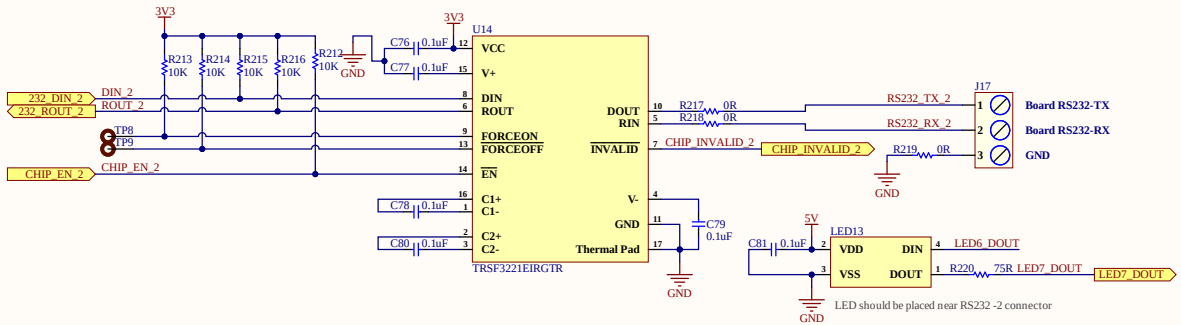
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Date: 22-Jan-25	Page Num: 8
Drawn By: Huzaifa Ali	Page Total: 15

RS232 - 1



RS232 - 2

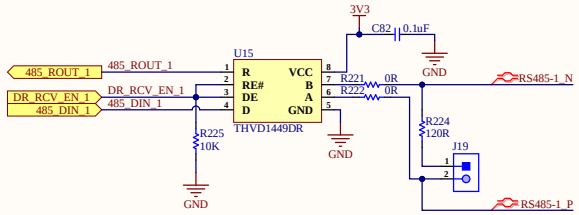


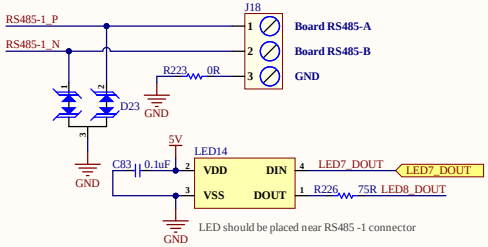


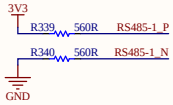
eSpark Consulting Group

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Date: 22-Jan-25	Page Num: 9
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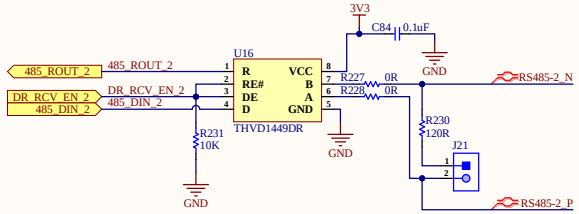
RS485 - 1

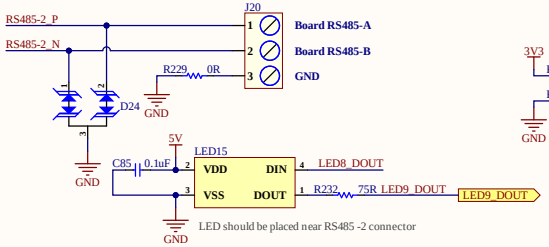


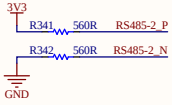




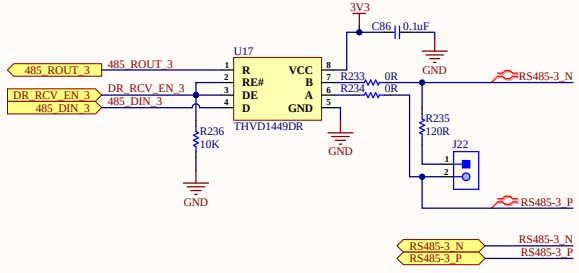
RS485 - 2



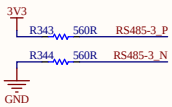




RS485 - 3





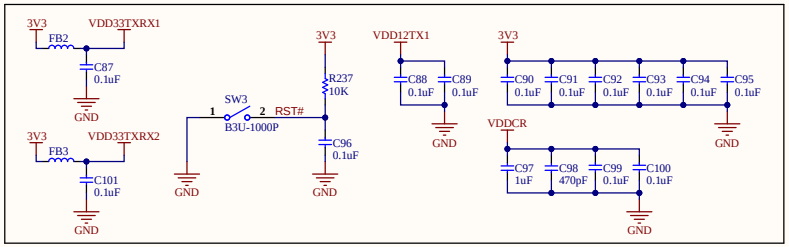


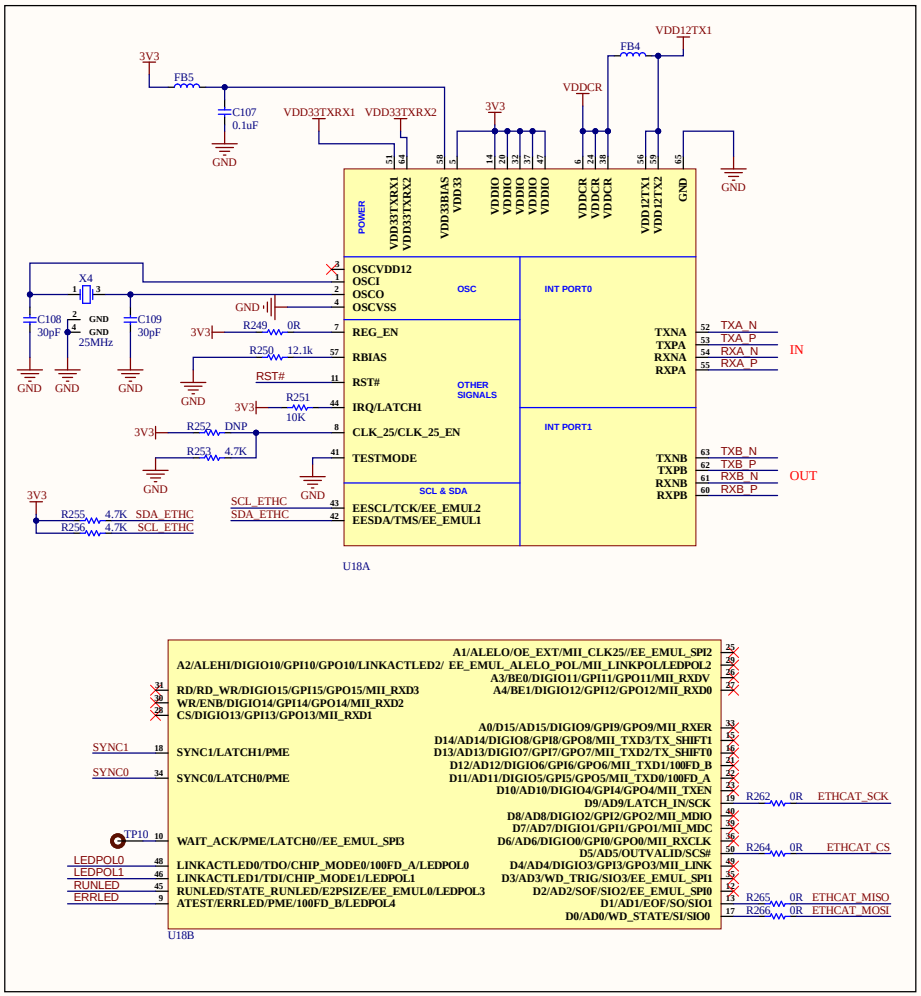


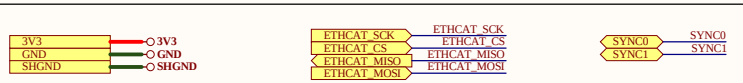
eSpark Consulting Group

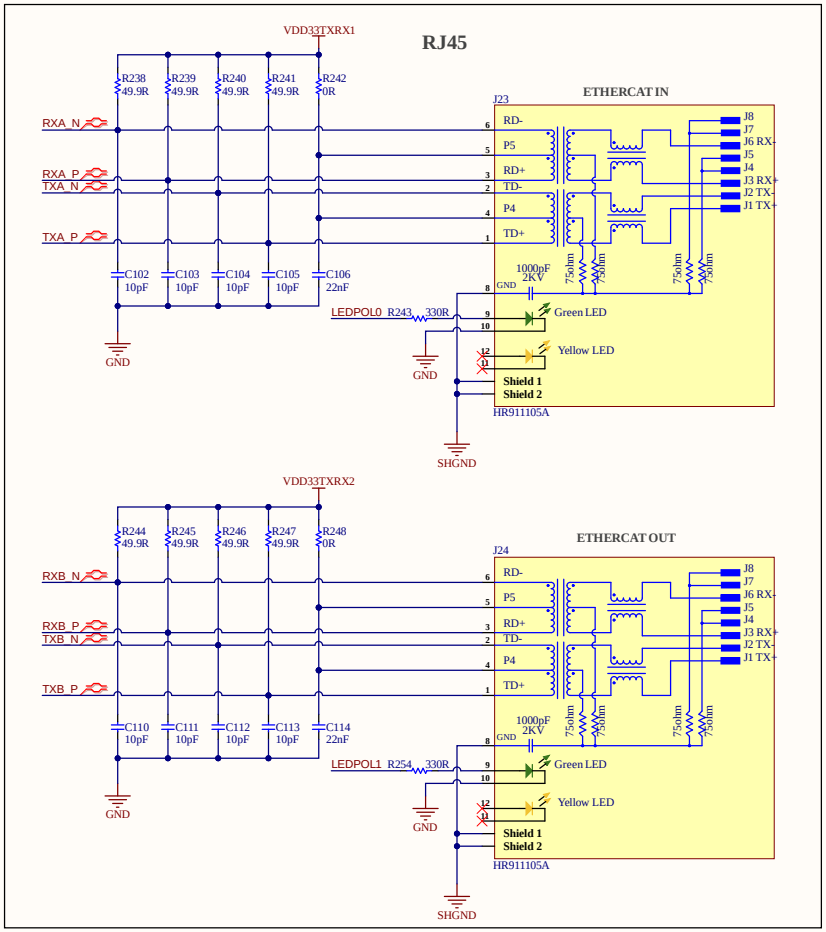
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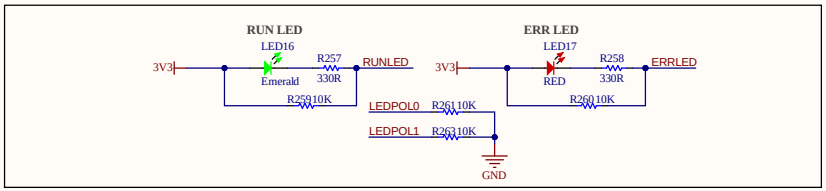
ETHERCAT

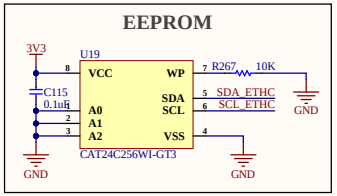








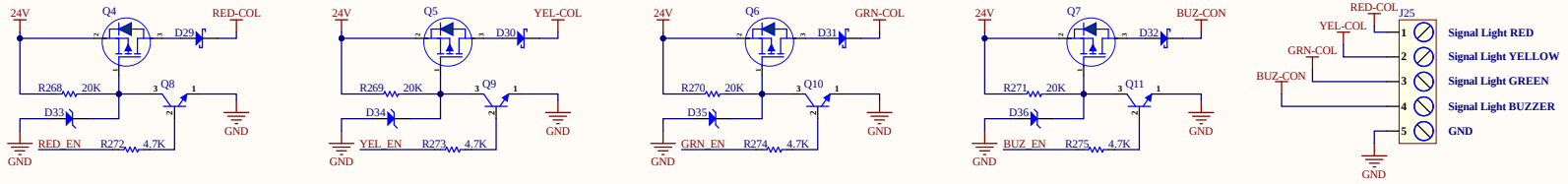




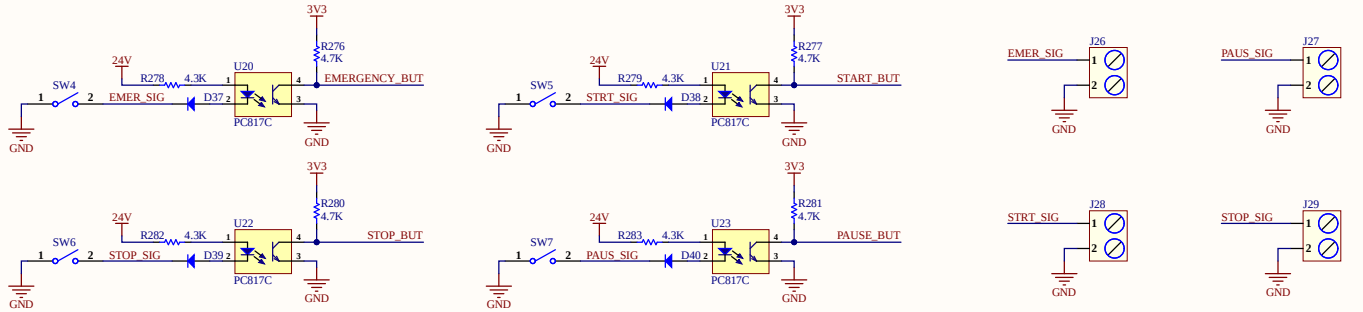
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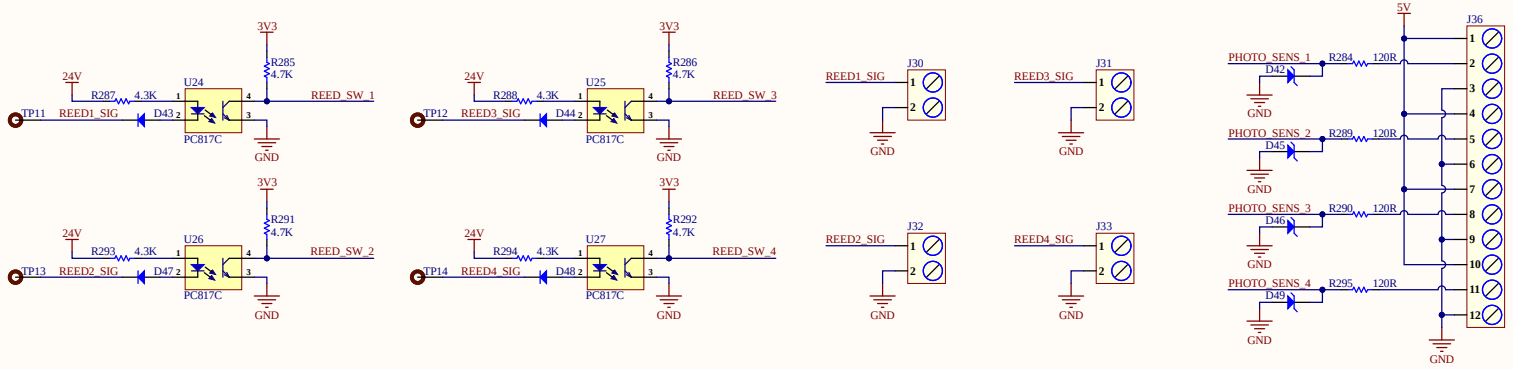
SIGNAL LIGHTS

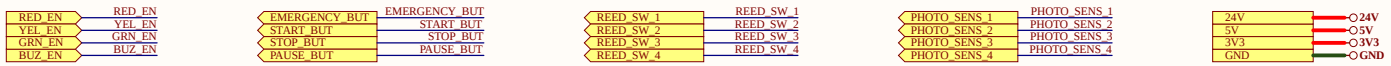


CONTROL BUTTONS



REED AND PHOTOELECTRIC SENSOR

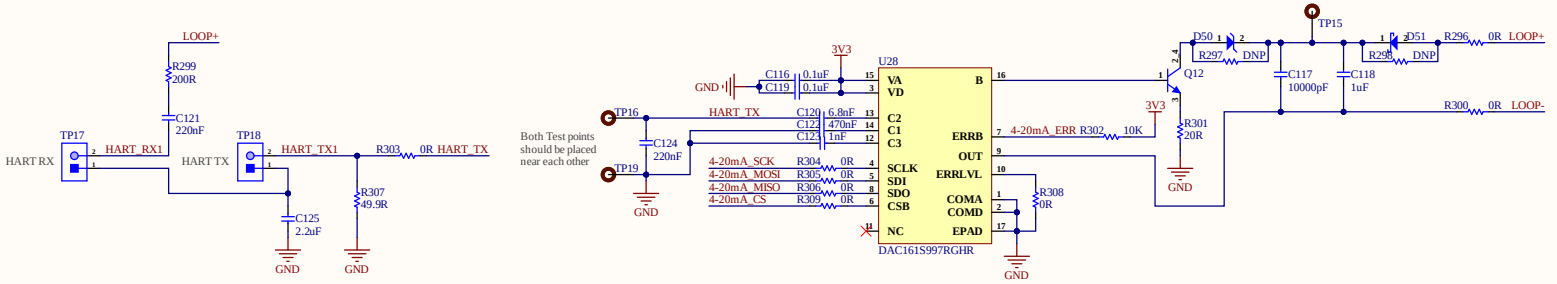




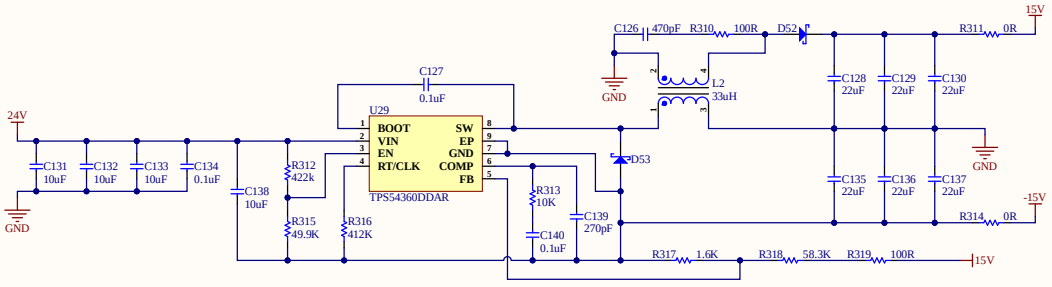
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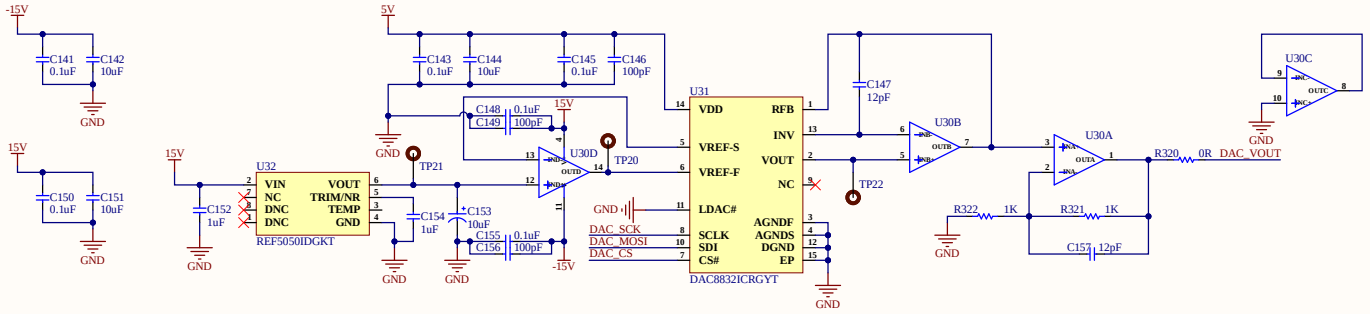
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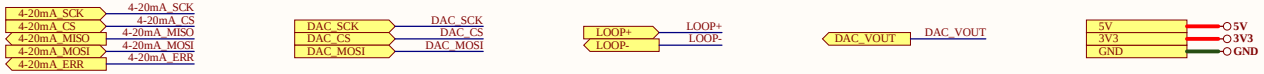


±15V POWER SUPPLY FOR DAC



±10V DAC OUTPUT

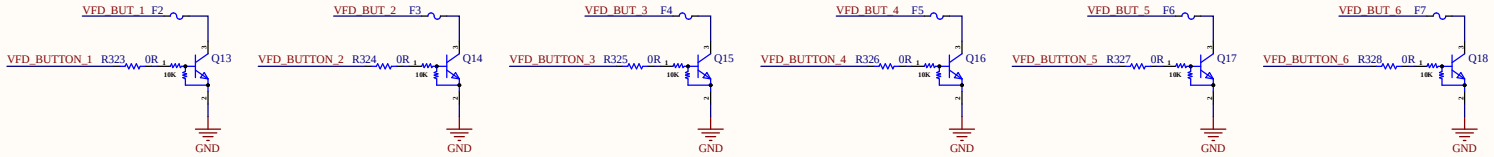




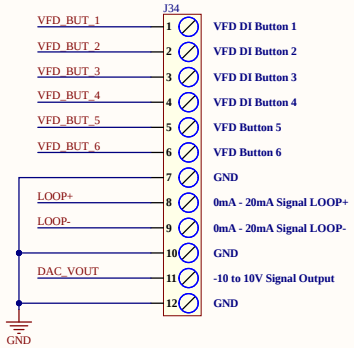
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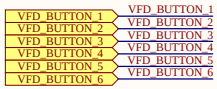
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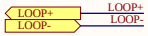
VFD DIGITAL INPUTS SUPPORT

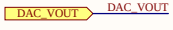


CONNECTOR







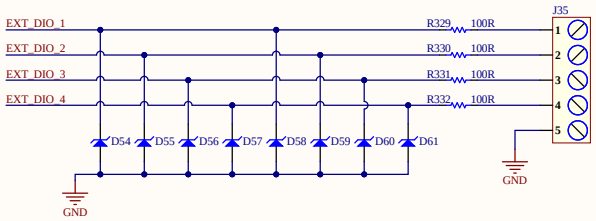




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EXTRA DIOs



EXT_DIO_1 EXT_DIO_1
EXT_DIO_2 EXT_DIO_2
EXT_DIO_3 EXT_DIO_3
EXT_DIO_4 EXT_DIO_4

GND GND

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